



Research Paper

A transfer learning framework for critical heat flux prediction pretrained on the 2006 lookup table

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ABSTRACT

The critical heat flux (CHF) represents the thermal safety limit in two-phase systems. Exceeding this threshold can lead to a boiling crisis. Despite advancements in mechanistic understanding, the intricate nature of bubble dynamics continues to impede the development of reliable and robust CHF prediction tools. Recent progress in machine learning has shown its potential as a powerful tool for modeling nonlinear phenomena. This study presents a transfer learning framework that encodes knowledge from a prior model directly into the parameter space of a neural network. Diverging from conventional hybrid frameworks which learn the deviation from a prior model's predictions, our method establishes a pretrained initial state, which is then fine-tuned with application specific data. The proposed framework was rigorously benchmarked against a hybrid framework and a standard multilayer perceptron (MLP) on a consolidated database of 1637 tube-flow CHF data points from six independent sources and an additional 768 proprietary measurements from 5×5 fuel assemblies. Our framework demonstrated accelerated convergence, accurate interpolation, and robust pressure-extrapolation. Interpretability analysis indicates that the enhanced generalization is attributable to the pretrained weights preserving physically meaningful patterns, thereby establishing a robust initial foundation for learning. Furthermore, window-type extrapolation mapping identifies the optimal experimental domains for future data acquisition. This work proposes a paradigm shift from deviation-correction to a pretraining framework, establishing a foundational and generalizable tool for next-generation digital twins in thermal-hydraulic safety analysis.

1. Introduction

Critical Heat Flux (CHF) is a crucial thermohydraulic threshold in numerous high-power thermal systems, including nuclear reactors, electronic cooling devices, vapor chambers, and industrial machinery. As the applied heat flux approaches the CHF limit, nucleate boiling—a highly efficient heat transfer mechanism characterized by bubble nucleation, growth, and departure—undergoes an abrupt transition to inefficient film boiling or the dry-out of the liquid layer. This abrupt transition leads to a rapid decline in local heat transfer coefficients, accompanied by a sharp increase in wall temperature, which can pose significant risks to system stability and material integrity. Therefore, preventing boiling crises is essential for ensuring the safe and reliable operation of two-phase thermodynamic systems.

In recent decades, various strategies have been employed to model

and predict CHF. Mechanistic models, grounded in hypotheses such as bubble crowding [1], liquid sublayer dry-out [2], and percolation theories [3,4], have contributed to a qualitative understanding of the triggers of CHF. However, these models remain fundamentally constrained by two major issues: the extreme difficulty in experimentally resolving the spatiotemporal scales relevant to bubble dynamics near CHF [5,6], and the current inadequacy of mathematical representations of multiscale phase-change interactions [7,8]. As a result, while offering valuable physical insight, such models lack the predictive generality required for broad application. Parallel to these efforts, empirical correlations have been developed from experimental data, serving as engineering tools for specific geometries, fluids, and flow regimes. Nevertheless, the dependence of these correlations on a restricted range of conditions significantly restricts their applicability when attempting to extrapolate to novel configurations or operational conditions.

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Another extensively investigated pathway involves multiphase computational fluid dynamics (CFD), which theoretically offers predictive capabilities independent of geometry and fluid properties by numerically solving the governing equations for two-phase flow and heat transfer. Despite advancements in interfacial closure laws [9] and phase-change modeling [10,11], its application encounters several significant challenges. One of the fundamental challenges in CFD simulation is the absence of a unified, physically consistent, and numerically robust criterion for CHF [12]. Different studies employ various metrics, including sudden changes in wall temperature [13,14], deviations from the linearity of boiling curves [15,16], or thresholds based on void fraction [17,18] or evaporation heat flux [19]. This absence of a universally accepted physical definition not only complicates cross-study comparisons but also undermines the consistency and reliability of CFD methods for CHF prediction. Furthermore, the predictive accuracy of CFD remains heavily dependent on the empirical calibration of parameters within bubble dynamics and wall boiling models [18], which undermines the promise of a first-principles approach. Compounding these methodological limitations, the computational costs are prohibitively high, making it challenging for CFD to serve as a reliable or standalone tool for digital twin control systems across diverse operational domains.

Machine learning (ML) has recently become a powerful tool for CHF prediction by efficiently capturing complex nonlinear relationships directly from data. This eliminates the need for detailed bubble-scale physical modeling, thereby addressing a key limitation inherent in mechanistic modeling. Initially, ML was used not only to optimize empirical correlations but also to gain new physical insights from data. Pioneering work, for instance, leveraged ML to decode the complex influence of the heated length-to-diameter (L/D) ratio on CHF, leading to the development of a generalized correction method [20]. This represented a significant step beyond simple regression and interpolation, showcasing ML's capacity to address longstanding phenomenological challenges in CHF modeling.

The consolidation of extensive, multi-source CHF databases [21] and the implementation of advanced ML architectures have paved new pathways for predictive modeling. Convolutional neural networks (CNN), multilayer perceptions (MLP), and transformer networks have all demonstrated strong predictive accuracy in multi-source CHF datasets [22,23]. Notably, transformers equipped with self-attention mechanisms have set new performance benchmarks by capturing complex, high-dimensional feature interactions and outperforming conventional models such as CNN and MLP.

Beyond architectural innovations, complementary strategies have further improved the robustness and physical plausibility of ML approaches for CHF prediction. These include ensemble learning to enhance generalization across disparate datasets [24,25] and feature engineering that incorporates physically relevant dimensionless parameters [26,27]. The adaptability of ML is further demonstrated by its successful application in specialized environments, such as predicting CHF in microgravity using MLP architectures [28], underscoring its potential for tackling unique operational challenges. Collectively, these advances highlight ML's growing versatility and efficacy in addressing the complexities of CHF prediction.

Despite these significant advancements, purely data-driven ML approaches exhibit inherent limitations. Their predictions may sometimes lack physical consistency, struggle to extrapolate reliably beyond the training data distribution, and tend to function as “black boxes”, providing limited insight into the underlying physical mechanisms that govern CHF.

Hybrid frameworks have been developed to address the interpretability challenges commonly associated with data-driven models. By leveraging deviations between established prior predictions and experimental observations [12,29,30], these hybrid frameworks have been developed to enforce partial consistency with the prior while enhancing model transparency. This deviation-correction strategy has

demonstrated remarkable success in reconciling physical principles with empirical data, thereby improving the reliability of CHF predictions in complex systems [31–33].

However, a fundamental challenge for both purely data-driven and existing hybrid approaches lies in their dependency on substantial amounts of high-quality experimental data for each new application domain. While hybrid frameworks integrate prior knowledge, they often require retraining with extensive target-domain data to learn the deviation function effectively. Given that CHF experiments are scarce, expensive, and hazardous, especially under extreme conditions, this dependency on large, new datasets for each target regime remains a practical constraint. These challenges motivate the exploration of strategies that can more efficiently leverage existing knowledge to reduce the demand for new experimental data. In this work, we propose a framework aimed at reducing the requirement for additional domain-specific data by leveraging transfer learning from a well-established prior method.

This dilemma highlights the need for a paradigm shift—moving from the incremental correction of prior models with data to a more foundational integration of knowledge and machine learning. We implement this shift through a novel transfer learning framework, which leverages the 2006 CHF lookup table (2006 LUT) as a comprehensive source of prior knowledge. As illustrated in Fig. 1, our approach conceptualizes CHF modeling as a synergistic merger between a data-abundant source domain and an empirical, data-scarce target domain. The framework first pretrains networks on simulated trends from the 2006 LUT to instill fundamental dependencies into their weights, and subsequently fine-tunes them on limited experimental data to enhance predictive realism and adaptability. This strategy aims to preserve the extrapolative power of the prior approach while leveraging the accuracy and flexibility of data-driven methods.

To rigorously assess the proposed framework, we benchmark it against two baselines: a conventional hybrid framework and a standard MLP utilizing Kaiming initialization. Evaluations are conducted on a consolidated database comprising 1637 CHF data points from six independent experimental sources for tube flows, as well as an additional 768 proprietary measurements from 5×5 fuel assemblies to assess generalization to complex geometries. We systematically evaluate the framework across three key aspects: interpolation accuracy, extrapolation capability, and model interpretability. Furthermore, we introduce a window-type extrapolation mapping technique to quantitatively assess prediction reliability across unexplored operational domains, thereby providing actionable insights for optimizing future data acquisition strategies.

In summary, this work makes three primary contributions: First, from a methodological perspective, we introduce a pretraining framework that represents a paradigm shift from deviation-correction to pretrained initialization. Second, through rigorous empirical benchmarking, we demonstrate that this approach achieves superior interpolation and, critically, significantly enhanced extrapolation capabilities under data-sparse and extreme conditions compared to conventional baselines. Finally, from a practical standpoint, the framework not only delivers accurate predictions but also enables interpretability analysis and extrapolation mapping, thereby providing scientifically grounded guidance for experimental design and resource allocation.

2. Methodology

2.1. CHF dataset and preprocessing

The source domain for pretraining was constructed by applying the 2006 LUT as detailed in Section 2.3 under the same experimental conditions used for its development. The underlying database for the LUT predominantly consists of experimental data from stainless steel tubes, with a minority of tests conducted on Inconel alloy surfaces. This process

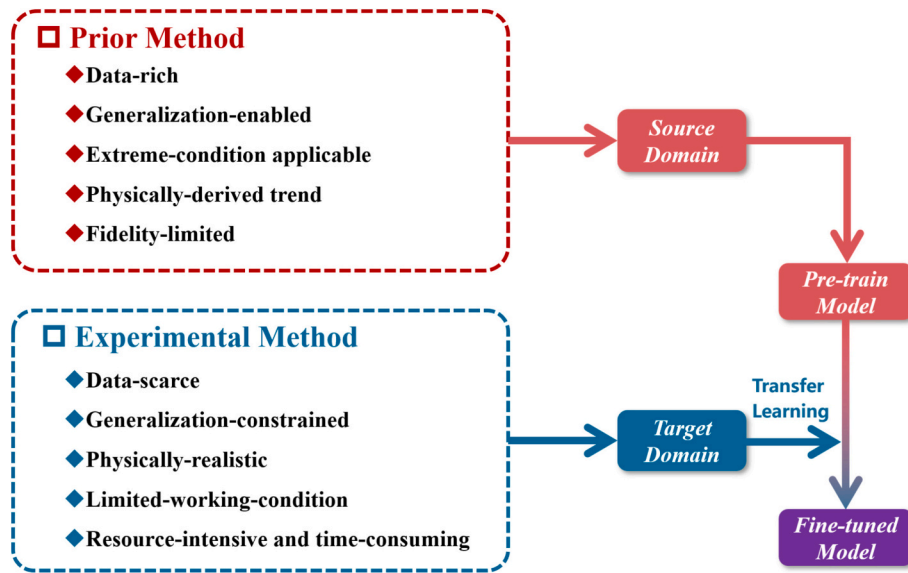


Fig. 1. Schematic of the proposed transfer framework: generalizable, the data-rich 2006 LUT is leveraged for network pretraining (source domain), followed by fine-tuning with scarce but high-fidelity experimental data (target domain).

involved computing CHF values for given input conditions through the standard inlet-condition workflow, applying interpolation and diameter correction. Crucially, this simulated dataset contains only LUT-generated values and is entirely separate from all subsequent experimental data, ensuring a clean separation between the pretraining (source domain) and fine-tuning (target domain) phases and preventing any potential data leakage.

The primary dataset employed in this study consolidates the CHF measurements from several independent experimental sources, covering a wide range of operating conditions in water and water-equivalent systems. Table 1 provides a detailed summary of the parameter ranges and the number of data points collected from each source. The heater materials employed in these experimental datasets (primarily stainless steel, as detailed in Table 1) are consistent with those that constitute the majority of the 2006 LUT's underlying database. The consolidated database contained 1637 data points representing a wide variety of thermohydraulic and geometric conditions. The conditions included pressures ranging from 0.1 to 20 MPa, mass fluxes from 332 to 7975 kg/(m²·s), inlet subcooling from 0.2 to 337 K, tube diameters from 0.001 to 0.0375 m, and heated lengths from 0.025 to 3.048 m.

To mitigate the scarcity of high-pressure CHF measurements for water, we incorporated the dataset of Guo [34], originally obtained using R134a, which was transformed into equivalent water CHF values via Katto modeling criterion [40]. This transformation enables the extension of the CHF coverage into pressure regimes, where direct experimental data are limited. All detailed datasets are provided in the *Supplementary Material* for transparency and reproducibility.

All input features and target CHF values were normalized to zero mean and unit variance to stabilize the training process and accelerate

convergence. Fig. 2 illustrates the statistical distributions of all input features and their pairwise relationships using a scatter plot matrix combined with kernel density estimates. The diagonal cells present the marginal distributions of each variable, visualized using kernel density curves, which capture the distribution of data across the ranges of diameter, length, pressure, mass flux, inlet subcooling, and CHF. The off-diagonal cells display scatter plots of all pairs of variables, providing an intuitive visualization of the joint distributions. For example, clear positive correlations can be observed between CHF and mass flux, whereas geometric parameters such as diameter and length display significantly weaker associations with CHF. This matrix not only highlights the diversity and range of the dataset but also reveals the presence of complex nonlinear interdependencies among key thermohydraulic parameters, underscoring the need for advanced modeling frameworks capable of capturing such intricate relationships.

To rigorously evaluate the generalization capability of the proposed framework to complex geometries, the second database was employed. This database comprises 768 proprietary, in-house CHF measurements for 5 × 5 full-length, electrically heated fuel assemblies, acquired from a thermal-hydraulic test facility at the Nuclear Power Institute of China (NPIC). The fuel assembly claddings employ metallic surfaces with an industrial-grade surface finish. The selected geometry presents significant physical complexities, including increased turbulence and heat transfer due to mixing vanes, as well as cold-wall effects from unheated structural components. These factors systematically alter the CHF mechanism when compared to simpler tube configurations.

A two-component Principal Component Analysis (PCA) was conducted on the input parameter space, as illustrated in Fig. 3. The operating conditions of the tube database generally fall outside the range

Table 1
Summary of the CHF experimental database and parameter ranges.

Author	Feature #1 pressure [MPa]	Feature #2 mass flux [kg/(m ² ·s)]	Feature #3 subcooling [K]	Feature #4 tube diameter [m]	Feature #5 heat length [m]	Heater material	No. of data
Guo [34]	7.73 to 15.75	1507 to 5033	19 to 168	0.008	0.24 to 1.6	316 stainless steel	419
Inasaka [35]	0.33 to 0.64	4300 to 6400	73 to 135	0.003	0.1	Info not supplied	4
Peskov [36]	10 to 20	750 to 5361	4.7 to 205	0.01	0.4 to 1.65	Stainless steel	16
Thompson [37]	0.1 to 18.96	542 to 7975	0.2 to 337	0.001 to 0.0375	0.025 to 3.048	Stainless steel	1080
Weatherhead [38]	13.79	332 to 2061	2 to 299	0.0111	0.457	Stainless steel, Nickel	67
Williams [39]	5.52 to 15.17	670 to 4684	21 to 251	0.0095	1.836	304 stainless steel	51
Total							1637

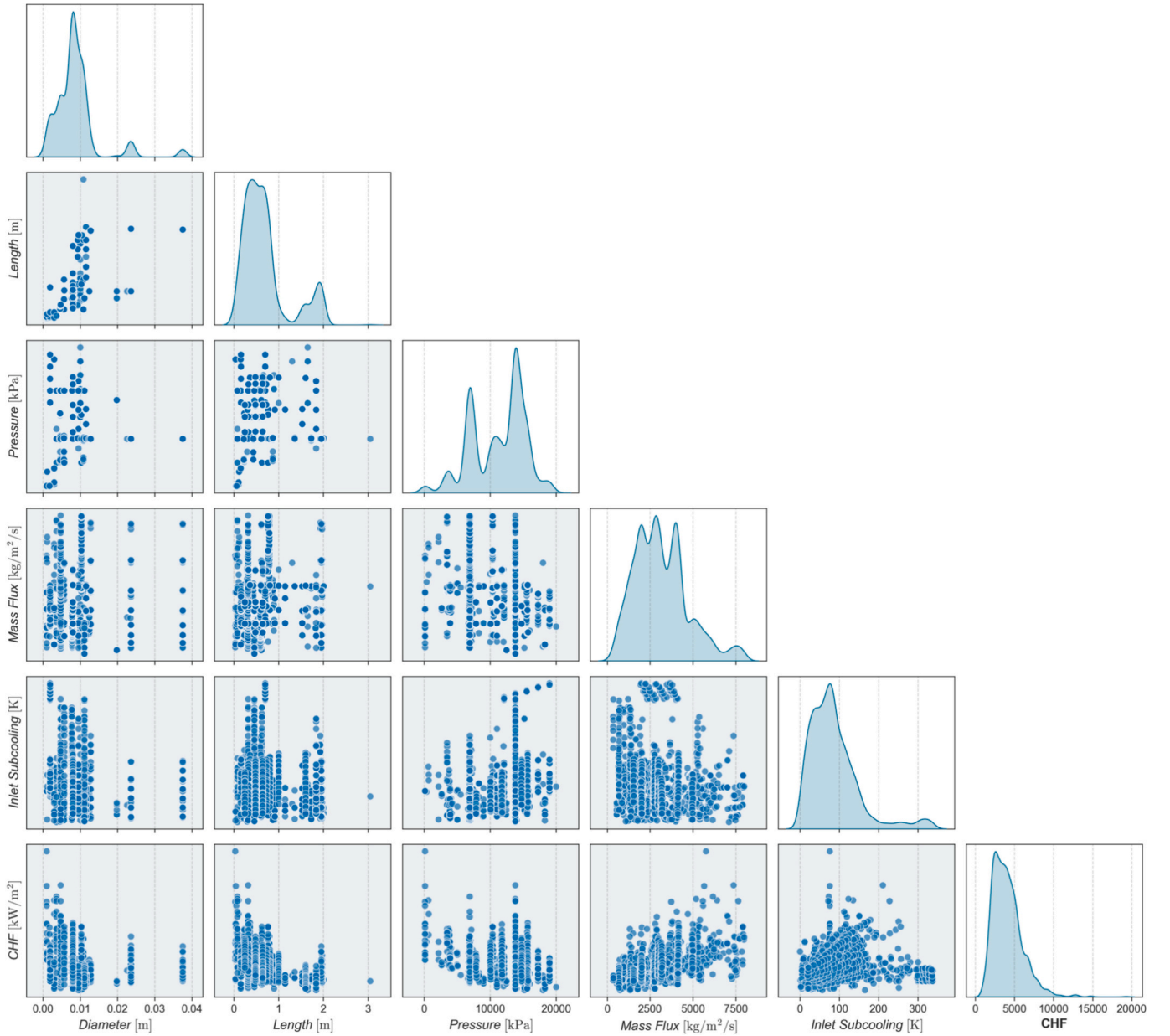


Fig. 2. Pairwise scatter plot matrix and marginal kernel density estimates for key thermohydraulic parameters in the CHF dataset.

covered by the source domain. In contrast, the operating conditions for the 5×5 fuel assembly are well within the domain of the LUT. The challenge here lies not in the inputs, but in the different underlying physics. The presence of complex geometry, such as mixing vanes and unheated walls, modifies the CHF mechanism. This means that the CHF value for the same thermohydraulic conditions can be fundamentally different from that observed in a simple tube.

2.2. AI methods

To rigorously benchmark the proposed framework, three modeling strategies were implemented and systematically compared: a purely data-driven MLP model, a hybrid framework, and the proposed pre-training framework. The schematic workflows of these approaches are shown in Fig. 4-Fig. 6.

(a) MLP model

As a fundamental baseline, a standard MLP was developed to represent a purely data-driven approach to the problem. The schematic of the data-driven MLP model is shown in Fig. 4. This model aims to capture the nonlinear input-output relationships in the experimental CHF dataset solely from data without any explicit incorporation of physical knowledge. The model uses the same input feature set as all subsequent approaches: pressure, mass flux, inlet subcooling, tube diameter, and heated length of the tube. Its architecture consists of an input layer, multiple fully connected hidden layers with nonlinear activation functions (ReLU), and a single output neuron that produces the CHF prediction. The network parameters were initialized using the Kaiming method following standard practices and trained by minimizing the mean squared error (MSE) between the predictions and experimentally measured CHF values.

The strength of this model lies in its capacity to capture complex, high-dimensional nonlinear relationships directly from data. However, its inherent limitations as a “black-box” model include a lack of built-in physical consistency and a potentially higher susceptibility to poor

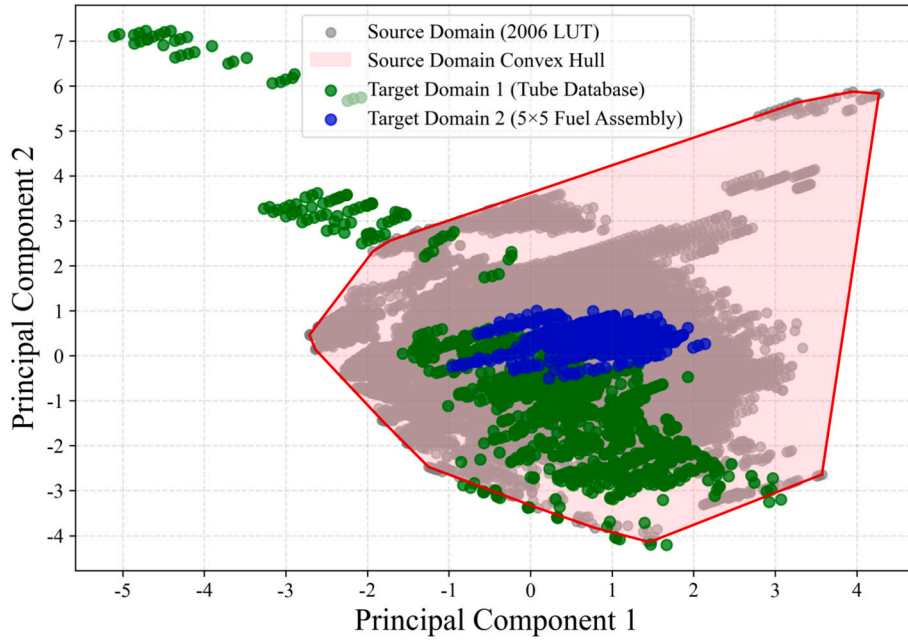


Fig. 3. Two-component PCA visualization with convex hull analysis of the CHF databases.

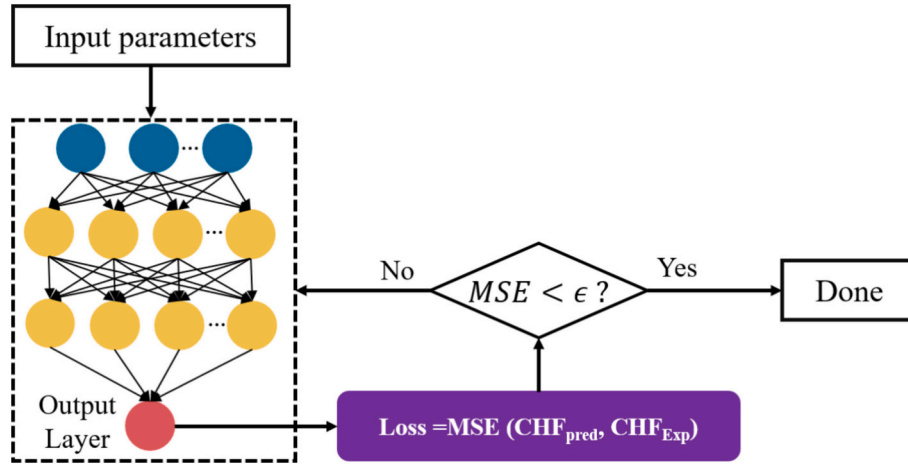


Fig. 4. Schematic of the data-driven MLP model used as a purely data-driven baseline.

generalization and unreliable extrapolation when operating outside the distribution of training data. Thus, its performance establishes a critical lower benchmark for the added value provided by incorporating prior knowledge.

(b) Hybrid framework

We implemented a commonly used hybrid framework based on the deviation-correction paradigm as a baseline. The schematic workflow of this approach is presented in Fig. 5. This method, which has been widely adopted in previous studies [29]– [33], operates by first calculating the CHF prediction of a prior physical model for a given set of input conditions.

As illustrated in Fig. 5, a neural network is then tasked with learning the deviation between the prior model and the experimental ground truth, rather than learning the CHF value directly. The loss function for the baseline model is formulated as:

$$\text{Loss} = \text{MSE}(\delta_{\text{pred}}, \text{CHF}_{\text{Exp}} - \text{CHF}_{\text{Prior model}}) \quad (1)$$

The final CHF prediction is obtained as follows:

$$\text{CHF}_{\text{pred}} = \delta_{\text{pred}} + \text{CHF}_{\text{Prior model}} \quad (2)$$

This model effectively integrates physical knowledge as a fixed base but requires the network to learn the full deviation function from the beginning for each new regime. Thus, its performance is inherently data-dependent, and its extrapolation capability is limited by the representativeness of the available training data.

(c) Proposed pretraining framework

The proposed pre-training framework is structured into two sequential phases: the pre-training and fine-tuning phases. This design aims to initialize the neural network with fundamental physical relationships before exposing it to experimental data. The overall architecture is shown in Fig. 6.

In the pre-training phase, the neural network is initialized and trained on a simulated CHF database derived from the LUT. The input parameters, such as pressure, mass flux, subcooling, tube diameter, and heated length, were mapped to the corresponding CHF predictions of the

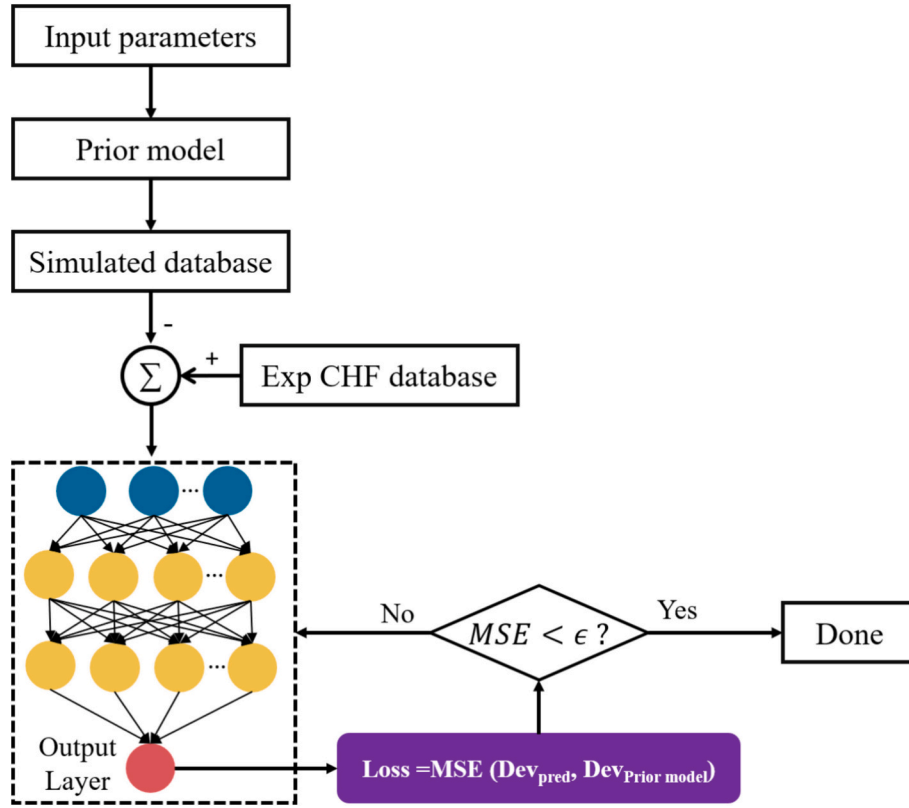


Fig. 5. Schematic of the hybrid framework, where a neural network learns the deviation between a prior physical model and experimental CHF.

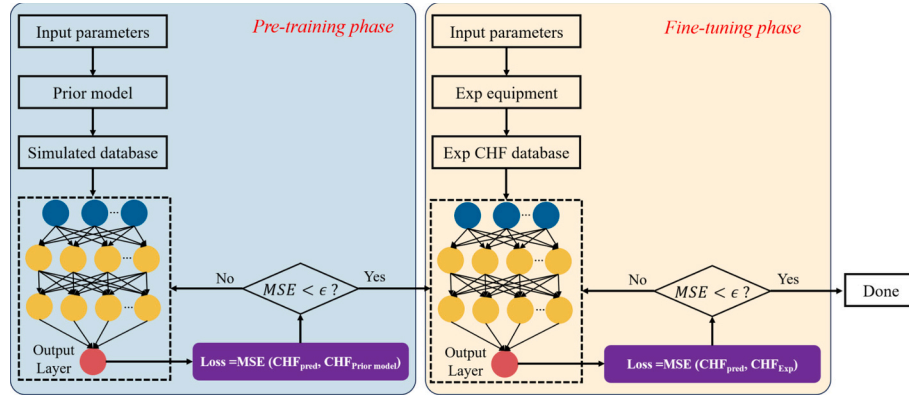


Fig. 6. Schematic of the proposed pretraining framework.

LUT. The training objective is formulated as the MSE between the network predictions and LUT-simulated CHF values:

$$Loss = MSE(CHF_{pred}, CHF_{Prior\ model}) \quad (3)$$

The optimization proceeds until the MSE converges below a pre-defined tolerance ϵ . At convergence, the network parameters encode the fundamental thermohydraulic dependencies represented in the prior model, thereby yielding pretrained weights that serve as a robust initialization for the subsequent fine-tuning stage. The output of this phase is a set of physics-primed network weights.

The pretrained model was then refined using the experimental CHF database. The same set of input features was employed, but the target values were the experimentally measured CHF values. The network architecture and physics-primed initialization were retained. The loss function is defined as the MSE between the predictions and experimental data:

$$Loss = MSE(CHF_{pred}, CHF_{Exp}) \quad (4)$$

This stage adjusts the physics-informed initialization to capture the nuanced nonlinearities and uncertainties that are inherent in application specific data. Training is terminated once the MSE converges below the threshold ϵ , ensuring that the final model balances physical consistency and empirical fidelity.

This two-stage strategy represents a departure from conventional hybrid methods. By embedding knowledge from the LUT directly into the network parameter space through pre-training, the framework is designed to establish a pretrained initial state, which is hypothesized to accelerate convergence and enhance extrapolation capability, particularly in data-scarce regimes. This approach aims to provide a more generalizable and foundational basis for the CHF modeling.

2.3. Prior model

As a generalized methodology for CHF prediction, the lookup table concept dates back to Doroshchuk [41], who assumed that the CHF for a given tube geometry can be expressed as a function of the local thermohydraulic state, namely, pressure, mass flux, and thermodynamic equilibrium quality. The 2006 revision of the LUT [42] represents a major advance: it consolidates a database exceeding 30,000 data points and tabulates CHF on a structured grid of 24 pressures (0.1–21 MPa), 20 mass fluxes (0–8000 kg/(m²·s)), and 23 qualities (−0.5 to 1.0). The revision improved data screening, removed outliers and duplicates, extended the coverage toward subcooled and limiting-quality regions, and enhanced the smoothness across the parameter space, thereby establishing a widely adopted benchmark for vertical water-cooled tubes.

Crucially, the LUT provides the CHF as a function of local conditions. In most applications, however, the inlet state is known, whereas the local quality at the CHF location is an outcome of the applied heat flux and must be determined iteratively using an energy balance. Our prior model implementation follows the standard inlet-condition procedure described in the reference [42] and is summarized by the workflow shown in Fig. 7.

For uniformly heated internal flow of a single-component fluid, the axial enthalpy rise satisfies:

$$\frac{dh}{dz} = \frac{\pi D q}{G(\pi D^2/4)} = \frac{4q}{GD} \quad (5)$$

which leads to the following expression for the local equilibrium quality at axial position z :

$$x_{(z)} = \frac{h_{in} - h_{sat}}{h_{fg}} + \frac{4q_{CHF}z}{GDh_{fg}} \quad (6)$$

Given values of pressure and mass flux, and an initial trial heat flux (typically starting from 500 kW/m²), the LUT is evaluated at (P, G, x) using trilinear interpolation to obtain q_{LUT} . A diameter correction is then applied for tubes in the 3–25 mm range:

$$q_{LUT,D} = \left(\frac{D}{8}\right)^{-0.5} q_{LUT} \quad (7)$$

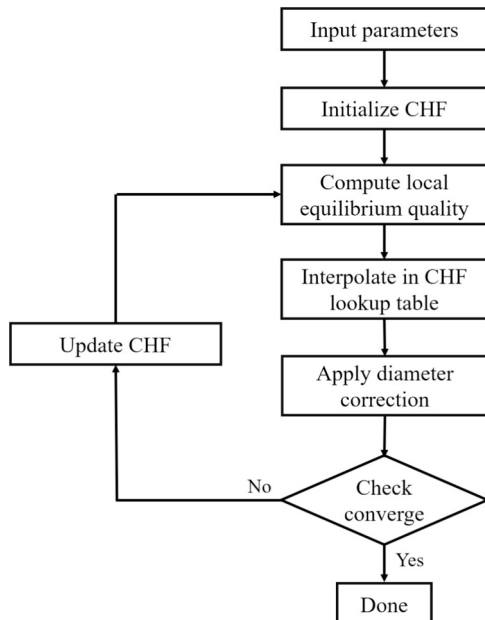


Fig. 7. Flowchart of the 2006 CHF lookup table.

When the CHF location is not explicitly tracked, we adopt the conservative assumption that CHF occurs at the heated-length exit. Starting from $q^{(0)} = 500 \text{ kW/m}^2$, the iterative procedure proceeds as follows:

1. Compute the exit quality:

$$x_{out} = \frac{h_{in} - h_{sat}}{h_{fg}} + \frac{4q_{CHF}L}{GDh_{fg}} \quad (8)$$

2. Obtain q_{LUT} through interpolation and apply diameter correction to get $q_{LUT,D}$.
3. Update the heat flux estimate for the next iteration using an under-relaxation method to ensure numerical stability:

$$q^{(k+1)} = (1 - \alpha)q^k + \alpha \hat{q}^k \quad (9)$$

where α is the relaxation factor. A value of $\alpha = 0.5$ is commonly used to start.

4. Iterate until the relative difference between successive estimates falls below a specified tolerance (10^{-3}):

$$\frac{|q^{(k+1)} - q^k|}{q^{(k+1)}} < \varepsilon \quad (10)$$

To ensure robustness, the LUT interpolations were constrained within the tabulated domain, with values outside this range excluded to prevent uncontrolled extrapolation. The resulting $q_{LUT,D}$ serves as the prior prediction employed to pre-train the network. This decision is informed by the LUT's extensive validation, transparent local-state dependence, and self-consistent inlet-condition workflow, which collectively offer a consistent and data-rich source domain for pre-training.

The 2006 CHF LUT was selected as the knowledge base for pre-training due to its widely accepted, comprehensive coverage, and empirically consistent thermohydraulic trends. While not a first-principles physical model, the LUT provides a robust representation of established CHF behavior across a vast operational envelope, making it ideally suited for transfer learning applications where capturing fundamental dependencies is more critical than absolute accuracy.

3. Results and discussion

This section establishes the experimental setup, training configuration, and evaluation metrics that form the foundation for the analyses in Sections 3.1–3.4. The comparative study includes three frameworks: a purely data-driven MLP, a deviation-based hybrid model, and the proposed pretraining framework. Section 3.1 assesses performance on the tube flow database, combining interpolation and pressure-extrapolation analyses. Section 3.2 extends the evaluation to 5×5 fuel assembly geometries to demonstrate generalization capability. Section 3.3 provides interpretability analysis of the learned representations, and Section 3.4 introduces extrapolation mapping to guide future data acquisition.

All implementations were performed in Python 3.12 using PyTorch as the deep-learning backend and scikit-learn for preprocessing, normalization, and statistical validation. Fixed random seeds and deterministic PyTorch operations were applied to enhance the reproducibility. The complete source code (Python files) for implementing all three models is provided in the *Supplementary Material*.

To ensure that performance differences are attributable solely to methodological innovations, all models utilize an identical multilayer perceptron backbone, incorporating the same input feature set, which includes pressure, mass flux, inlet subcooling, tube diameter, and heated length, with a single CHF output. The three frameworks differ exclusively in their initialization and training strategies: Kaiming

initialization is employed for the baseline MLP, deviation learning is utilized for the hybrid model, and pretraining followed by fine-tuning is applied for the present method. The hyperparameters were determined through Bayesian optimization, with the selected values summarized in Table 2.

Training was conducted utilizing the Adam optimizer, with the initial learning rate determined through a Bayesian search. Early stopping was implemented after 200 epochs without improvement in validation loss, and a learning-rate scheduler was employed. Batch normalization and input standardization were consistently applied to stabilize the training process.

The performance of the model was quantitatively evaluated using four established metrics: Root Mean Square Percentage Error (RMSPE), Relative Root Mean Square Error (rRMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2). The metrics are defined as follows:

$$\text{RMSPE} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left(\frac{\text{CHF}_{\text{pred}}^{(i)} - \text{CHF}_{\text{exp}}^{(i)}}{\text{CHF}_{\text{exp}}^{(i)}} \right)^2} \times 100\% \quad (11)$$

$$\text{rRMSE} = \frac{\sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left(\text{CHF}_{\text{pred}}^{(i)} - \text{CHF}_{\text{exp}}^{(i)} \right)^2}}{\text{CHF}_{\text{exp}}^{(i)}} \quad (12)$$

$$\text{MAPE} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left| \frac{\text{CHF}_{\text{pred}}^{(i)} - \text{CHF}_{\text{exp}}^{(i)}}{\text{CHF}_{\text{exp}}^{(i)}} \right| \times 100\% \quad (13)$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N_{\text{test}}} \left(\text{CHF}_{\text{exp}}^{(i)} - \text{CHF}_{\text{pred}}^{(i)} \right)^2}{\sum_{i=1}^{N_{\text{test}}} \left(\text{CHF}_{\text{exp}}^{(i)} - \overline{\text{CHF}}_{\text{exp}} \right)^2} \quad (14)$$

3.1. Model performance on TUBE-flow dataset

To assess the interpolation within the measured operating envelope, the consolidated tube CHF database was randomly split into training (60 %), validation (20 %), and test (20 %) subsets. During the pre-training phase of the proposed pretraining framework, the synthetic dataset generated from the LUT prior was partitioned into 80 % pre-training and 20 % validation subsets, respectively. All input features and target CHF values were standardized to a zero mean and unit variance using statistics derived from the corresponding training sets to stabilize the optimization. For physical interpretability, all error metrics reported below were assessed on the original (denormalized) CHF scale.

Fig. 8 illustrates the pointwise concordance between the predictions and experimental results for the test set using the trained model. The pretraining framework predictions exhibited the most compact clustering around the line of perfect agreement, followed by the hybrid model, purely data-driven MLP, and LUT prior. Quantitatively, the proposed pretraining framework achieved the lowest RMSPE of 6.79 %, surpassing the hybrid method baseline (7.47 %), MLP (9.36 %), and LUT (11.26 %). These findings suggest that initializing the network within a prior method parameter space results in consistently smaller

interpolation errors than learning either the deviation from or a direct mapping independent of the prior.

The cumulative distribution function (CDF) of the absolute relative error, as shown in Fig. 9, offers a more comprehensive evaluation of the prediction error distribution. The curve of the proposed pre-training framework is positioned closest to the top-left corner of the graph, indicating that it achieved the highest proportion of accurate predictions across all error thresholds. For example, at a given absolute relative error threshold, a larger fraction of the proposed framework predictions fell below this error value than those of the other models. This finding unequivocally demonstrates that the enhanced performance of the proposed framework is consistent and robust across the entire test dataset, rather than being merely a consequence of superior average performance.

The convergence behavior of each model is depicted in Fig. 10, which shows the MAPE of the validation set as a function of the training epochs. Owing to its initialization via pretraining on the LUT-simulated trends, the proposed model starts at a MAPE of approximately 8.5 % and rapidly decreases to approximately 5.5 % within approximately 20 epochs, after which it stabilizes. The hybrid model begins at a higher MAPE (≈ 11.2 %), reflecting the need to learn the full residual function between the LUT prediction and the experimental values, and converges to approximately 6.6 % after approximately 50 epochs. In contrast, the MLP initialized using the Kaiming method shows slower convergence, starting near 30.7 % MAPE and reaching approximately 13.1 % after 100 epochs, without evidence of instability but with considerably slower learning progress. These trajectories underscore the data efficiency and training stability afforded by the pretraining strategy, which accelerates convergence and leads to superior predictive accuracy.

To assess the models' capability for pressure extrapolation, a series of tests was devised wherein data below specified cutoff pressures were employed for training and validation, whereas data at higher pressures were reserved for testing. The pretraining phase utilizes the full pressure range of the LUT to construct a universal pretrained model. For each extrapolation scenario, only the fine-tuning stage is restricted to experimental data below the specified cutoff pressure. The test set consists of experimental data above cutoff pressure. Cutoff values of 4, 8, 12, and 14 MPa were selected, with the corresponding data partitions listed in Table 3.

The proposed pre-training framework consistently exhibits superior extrapolation performance across all cutoff pressures. In the most challenging scenario, characterized by a pressure of 4 MPa, where only 83 samples were available for training and 1533 high-pressure samples for testing, the proposed method achieved an R^2 value of 0.9081, with 62.04 % of the predictions falling within a ± 10 % error margin. In contrast, both the hybrid method and MLP demonstrated significantly compromised performance, with R^2 values in the negative range and a minimal number of predictions meeting the ± 10 % error criterion.

The MLP demonstrated slightly, though not significantly, improved metrics at 4 MPa compared to 8 MPa, as evidenced by an R^2 of -6.56 versus -4.15 . This variation highlights the intrinsic instability of purely data-driven models when subjected to substantial distribution shifts, where neither configuration achieves a satisfactory extrapolation accuracy.

The CDF of the absolute relative error under each cutoff pressure provide a deeper insight into the distribution of model errors across the entire test set (Fig. 11). The CDF curve of the proposed method is consistently to the left and above those of the baseline models across all cutoff values, indicating not only a lower median error but, more importantly, a tighter and more physically consistent error distribution. This demonstrates that the performance improvement is uniform and robust across the entire prediction spectrum, not merely a result of improved average accuracy.

For instance, at $P_{\text{cutoff}} = 4$ MPa, the proposed framework maintains a steeply rising CDF curve, indicating that most predictions are concentrated within a low-error region. In contrast, the PIML and MLP models

Table 2
Optimized hyperparameter configuration obtained from Bayesian search.

Hyperparameter	Sampling Method	Best Value
The depth of hidden layers	Discrete uniform distribution	2
The number of hidden layer nodes	Discrete uniform distribution	256, 128
Initial learning rate	Continuous Uniform on Log Scale	$1e^{-5}$
Batch size	Discrete choice	128
Epochs	Discrete uniform distribution	1000

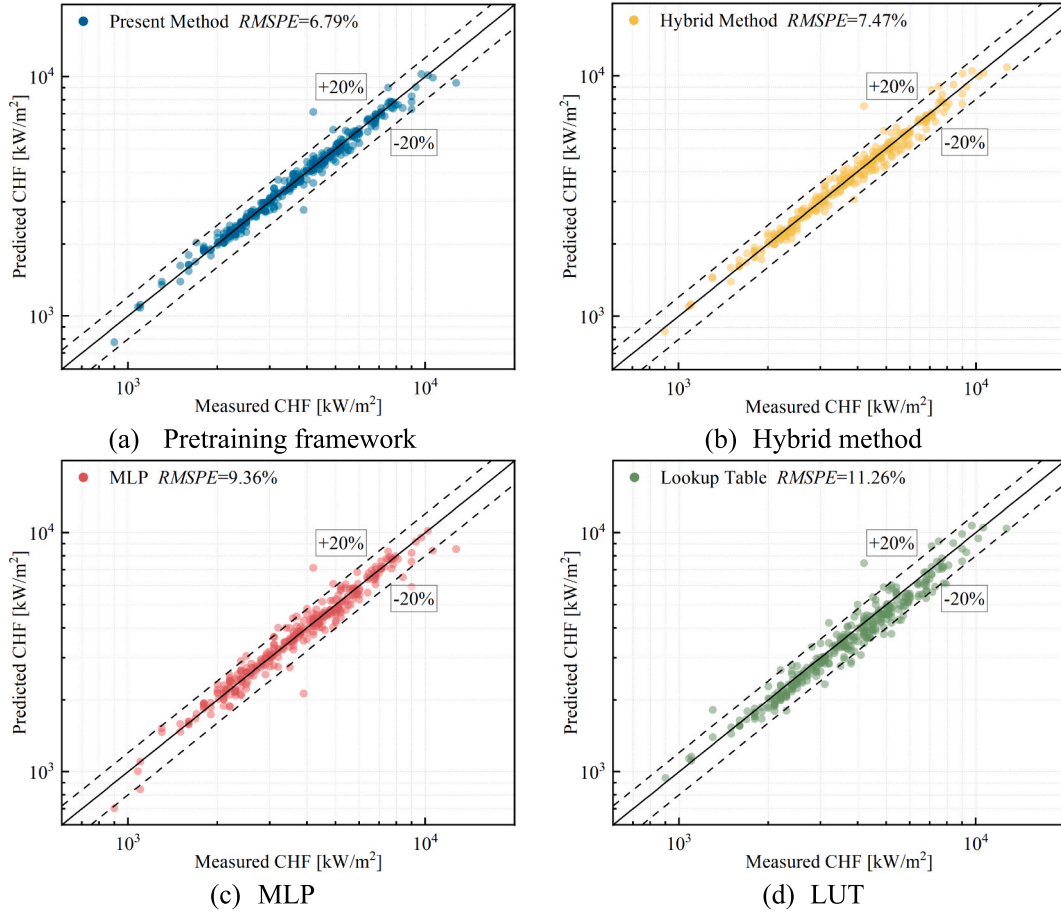


Fig. 8. Scatter plot of predicted versus experimental CHF values for the tube-flow test set.

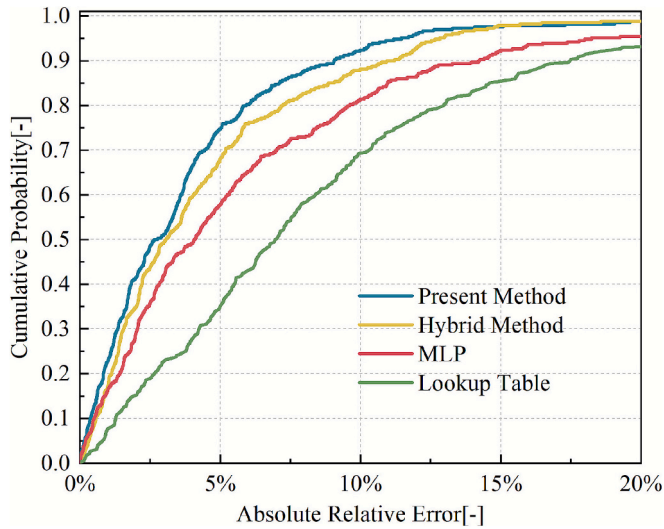


Fig. 9. Cumulative distribution of the absolute relative error for the tube flow test set predictions.

exhibited flatter curves, indicating a broader and less predictable error distribution. This behavior highlights a key advantage of physics-based pretraining: by encoding fundamental thermodynamic dependencies into the network weights, the model retains a generalized physical reasoning ability even under strong operational shifts.

To further investigate the extrapolation behavior, we established 200 test conditions with fixed median values for the mass flux, inlet

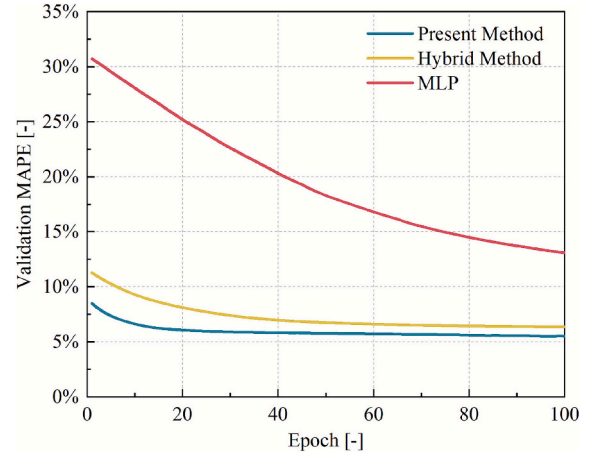


Fig. 10. Evolution of the validation set MAPE with training epochs for the tube flow dataset.

subcooling, length, and diameter, as derived from the database, while varying the pressure from 0.1 to 20 MPa. The predictions of each model trained under different cutoff pressures were compared with the LUT reference, which, although not quantitatively exact, reflected physically plausible trends. The resulting comparative trends are shown in Fig. 12.

At $P_{cutoff} = 4$ MPa, both the MLP and PIML models exhibited qualitatively incorrect trends, as their predictions increased with pressure beyond the training domain, contrary to the anticipated decrease indicated by the LUT. In contrast, the present method maintains the correct

Table 3

Extrapolation performance metrics on the tube flow dataset under different pressure cutoff values.

Cutoff Pressure P_{cutoff} [MPa]		4	8	12	14
No. of training data		83	409	644	1034
No. of test data		1533	1125	831	344
Test data R^2	Present Method	0.9081	0.9384	0.9632	0.9408
	Hybrid Method	−0.4481	0.7353	0.9302	0.9419
	MLP	−6.5573	−4.1485	0.1918	0.8351
	Present Method	0.1362	0.1183	0.0942	0.0989
Test data rRMSE	Hybrid Method	0.5407	0.2453	0.1297	0.0961
	MLP	1.2352	1.0817	0.4416	0.1651
	Present Method	9.75 %	9.38 %	6.47 %	5.64 %
	Hybrid Method	64.47 %	28.24 %	10.49 %	6.36 %
Test data MAPE	MLP	131.21 %	124.14 %	31.63 %	14.33 %
	Present Method	62.04 %	64.27 %	82.55 %	83.14 %
	Hybrid Method	8.02 %	27.64 %	59.81 %	84.30 %
	MLP	4.76 %	2.31 %	29.84 %	56.69 %

monotonic decline and aligns with the LUT trend, demonstrating that physics-primed initialization imparts a beneficial extrapolative bias to the network. Although the LUT is not quantitatively precise, its ability to reproduce the correct qualitative trend underscores its utility as a reference for physical plausibility in the model.

At $P_{cutoff} = 8$ MPa, all models accurately captured the negative correlation between CHF and pressure; however, the MLP and PIML curves were excessively steep, with the MLP alarmingly predicting negative CHF values beyond 10 MPa, whereas PIML became unphysical beyond 16 MPa. Such predictions are physically impossible and unacceptable in safety-critical applications.

At $P_{cutoff} = 12$ MPa, the PIML predictions nearly overlapped with those of the present method and LUT values. In contrast, the MLP continued to exhibit an unphysical negative CHF at extremely high pressures. Finally, at $P_{cutoff} = 14$ MPa, all three data-driven models converged to nearly identical and physically plausible trends, indicating that sufficient data can partially compensate for the absence of physical prior knowledge.

As the cutoff pressure increased and more training data became available, all the models demonstrated improved performance. Notably, at $P_{cutoff} = 14$ MPa, the PIML model converged toward the performance of the proposed method, suggesting that sufficient data can partially compensate for the lack of physical priming. Nevertheless, the present framework maintains superior extrapolation capability and consistency across the entire pressure range, affirming its value in data-sparse or high-extrapolation regimes.

Pretraining on a larger dataset generated by the 2006 LUT should be interpreted as a transfer learning strategy that leverages an empirically constructed prior rather than as a strictly physics-based extrapolation technique. The improved performance at pressures exceeding the cutoff demonstrates the model's ability to exploit this prior knowledge when experimental data are limited, rather than the enforcement of first-principles constraints.

3.2. Model performance on 5×5 fuel assembly dataset

To evaluate the generalizability of the proposed framework to geometrically complex systems, all models were assessed using the proprietary 5×5 fuel-assembly dataset. As discussed in Section 2.1, the main challenge arises from the divergent CHF mechanisms introduced

by features such as mixing vanes and unheated walls, even though the operating conditions remain within the input parameter space of the LUT.

Interpolation performance was examined using a random split of the 5×5 fuel assembly data. The scatter plot of predicted versus experimental CHF values shown in Fig. 13 demonstrates that the predictions from the proposed framework cluster most closely around the line of perfect agreement. Quantitatively, the 2006 LUT markedly underestimated the CHF values for this configuration, resulting in an RMSPE of 21.88 %. This systematic deviation illustrates the limitation of applying a tube-flow-based model directly to a geometry governed by fundamentally different physical mechanisms.

After fine-tuning with the experimental fuel assembly data, the proposed framework achieved a test-set RMSPE of 3.10 %, outperforming both the hybrid framework (3.34 %) and the standard MLP (4.38 %). Although the LUT does not provide quantitatively accurate predictions for this geometry, its incorporation during pretraining offers an advantageous initialization that facilitates more efficient learning and improved predictive accuracy once calibrated with target-domain data.

Furthermore, the cumulative distribution function of the absolute relative error in Fig. 14 indicates that the proposed framework yields a higher proportion of accurate predictions across all error thresholds compared with the baseline models.

The extrapolation capability with respect to pressure was evaluated by dividing the dataset into training and test sets according to cutoff pressures, with the results summarized in Table 4. The proposed framework consistently achieved the highest extrapolation accuracy across all cutoff pressures. In the most data-limited scenario ($P_{cutoff} = 10$ MPa), it attained $R^2 = 0.9560$ and $MAPE = 4.95$ %, with 86.81 % of the test samples falling within ± 10 % of the predicted values. Under the same conditions, the hybrid and MLP models exhibited larger errors and lower accuracy. As the amount of available training data increased with higher P_{cutoff} values, the performance of all models improved; however, the proposed framework retained a clear advantage across metrics.

To examine the physical plausibility of the extrapolated trends, model predictions were generated across a pressure range from 0.1 MPa to 20 MPa, while maintaining fixed values for other parameters based on the fuel assembly dataset (mass flux = 3550 kg/(m²·s), inlet subcooling = 50.31 K). The predicted trends were compared with experimental data at 15.6 MPa, as shown in Fig. 15.

At a cutoff pressure of 10 MPa, both the MLP and hybrid frameworks produced pressure-dependent trends that deviated from the experimental reference point, whereas the proposed framework generated a curve passing through the experimental data. As the cutoff pressure increased to 12 MPa and 14 MPa, predictions from all models gradually converged toward the reference value, with both the proposed and hybrid frameworks exhibiting physically consistent behavior. At 16 MPa, all three models provided comparable predictions that were physically reasonable and close to the experimental results.

Overall, these results indicate that the proposed framework offers improved reliability when extrapolating to pressure regimes beyond the range of the training data. Although the prior model alone struggles to accurately capture the CHF behavior for complex geometries, incorporating its insights through transfer learning significantly improves both model reliability and extrapolation performance, particularly in situations where experimental data is scarce.

3.3. Interpretability analysis

The pronounced differences in extrapolation accuracy across the three frameworks—particularly the consistent performance of the pre-training framework in the face of significant distribution shifts—motivated an interpretability analysis of what each model actually learns. We focus on how the learned parameters encode the thermohydraulic dependencies that govern CHF.

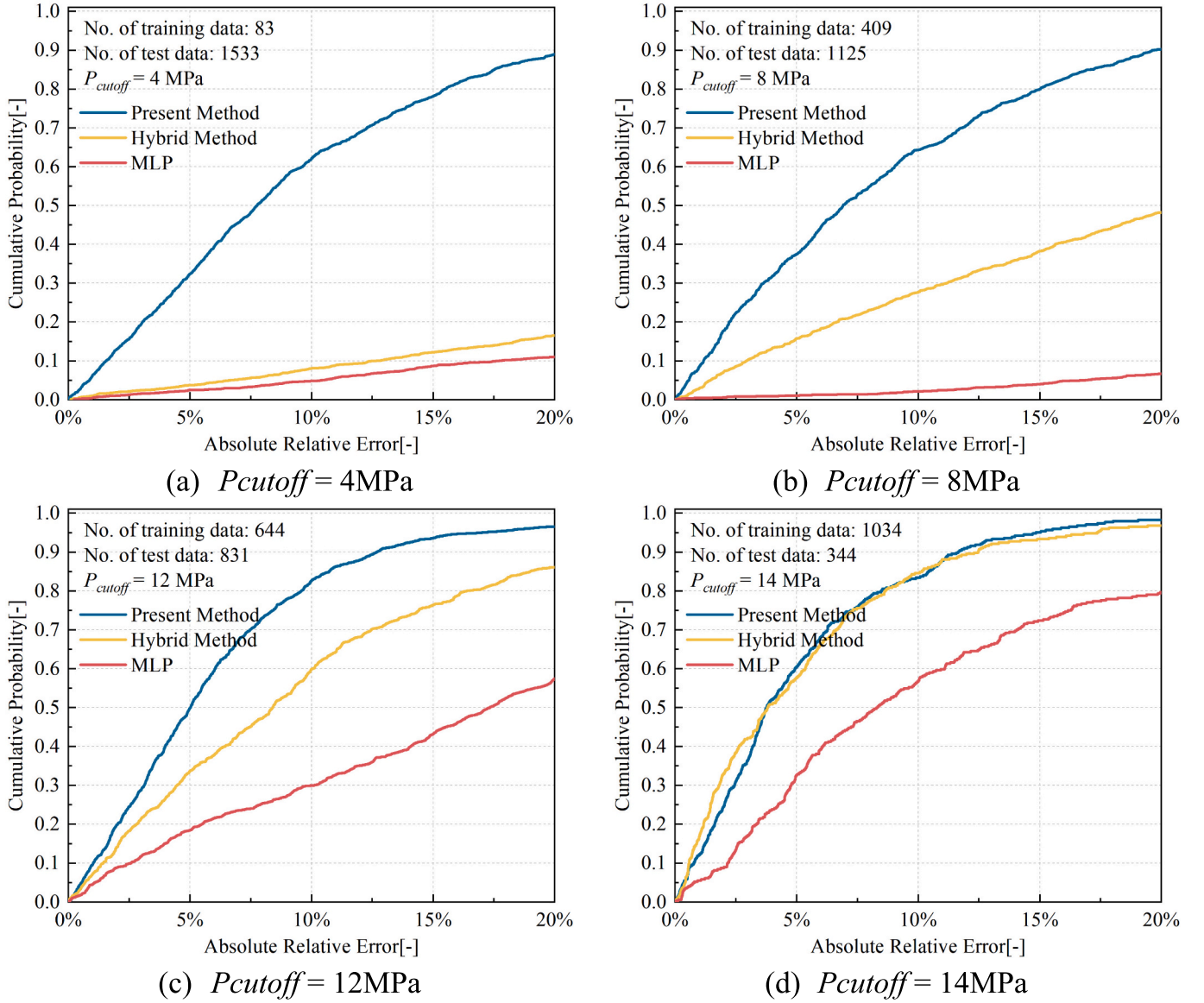


Fig. 11. Cumulative distribution of absolute relative errors for pressure-extrapolation cases on the tube flow dataset.

Standard MLP serve as purely data-driven approximations. Although initialized using the Kaiming method for enhanced training stability, the model must learn the entire high-dimensional nonlinear mapping from the input features (pressure, mass flux, inlet subcooling, tube diameter, and heated length) to the CHF from scratch. Although MLP can capture complex patterns with sufficient data, this approach can lead to overfitting, particularly in data-sparse situations, where noise and spurious correlations may become problematic. Crucially, without embedded physical constraints, the learned mapping may violate the fundamental thermodynamic principles during extrapolation. This is evident from the unphysical positive slopes and negative CHF predictions shown in Fig. 12.

The deviation-based hybrid framework incorporates physical priors by reformulating the learning objectives. Instead of estimating the absolute CHF, it learns the deviation between the predictions from a prior model and the experimental values:

$$\text{Bias} = \text{CHF}_{\text{Exp}} - \text{CHF}_{\text{LUT}} = g(\text{PG}\Delta T_{\text{sub}}\text{DL}) \quad (15)$$

This reformulation can be beneficial for interpolation, as the residual function $g(\cdot)$ typically covers a narrower numerical range and may exhibit smoother behavior compared to the direct CHF mapping. By

anchoring predictions to a baseline, this approach maintains a degree of plausibility, contributing to its superior interpolation performance compared to that of the MLP.

However, this strategy implicitly assumes that the prior errors are systematic, smooth, and learnable. If these biases vary irregularly across the parameter space or arise from different mechanisms, learning the deviations can be as challenging, or even more so, than learning the CHF directly. Our analysis of the LUT deviations supports this issue: the Q-Q plot in Fig. 16 reveals greater deviations from normality in the residuals compared to the raw CHF data. Furthermore, the linear fit's R^2 value decreased from 0.9070 to 0.8375, indicating a less regular and harder-to-model distribution. While R^2 on a Q-Q line serves only as an indicator of linear concordance, the trend aligns with the observed degradation of the hybrid framework in strong extrapolation ($P_{cutoff} = 4\text{ MPa}$), where the network struggles to learn an irregular deviation field due to insufficient data.

Our proposed pretraining framework addresses this fundamental limitation by shifting the locus of prior knowledge injection from the output space to the parameter space. Rather than simply learning deviations from a prior model, our approach distills the global thermo-hydraulic trends of the prior into the network weights by pretraining on

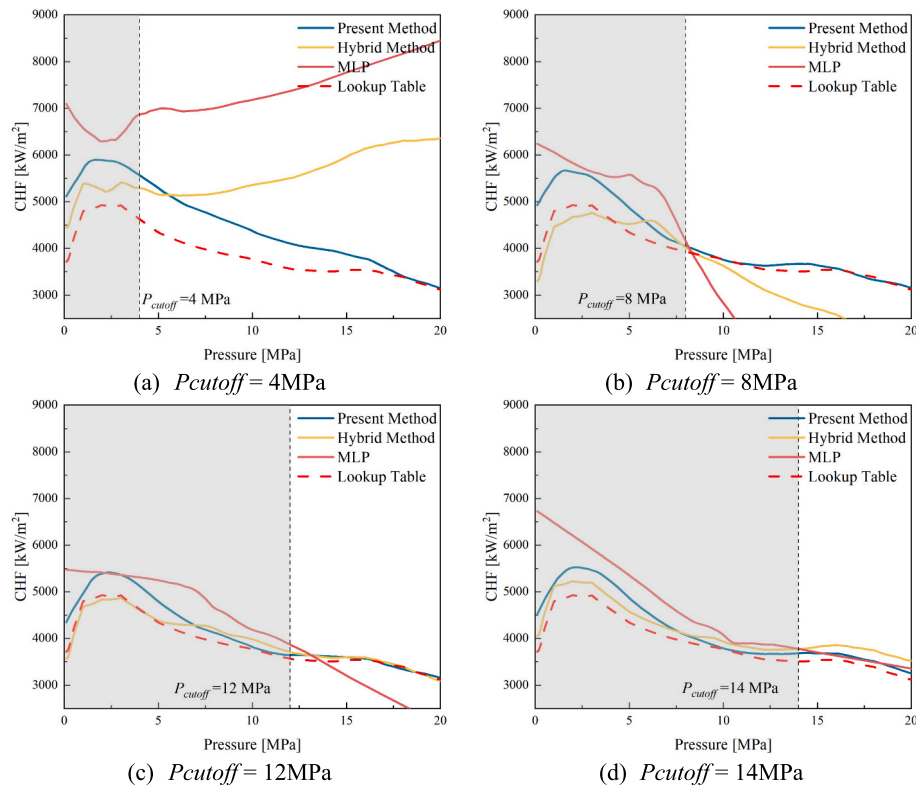


Fig. 12. Comparison of predicted CHF trends over pressure for the tube flow dataset under different training cutoff pressures.

a comprehensive synthetic dataset generated from the LUT. Subsequent fine-tuning with experimental data refines the pretrained parameterization. This strategy effectively constrains the optimization to a physically plausible subspace, stabilizing the dependencies learned for the critical input variables.

Two complementary analyses clarify this mechanism. First, an analysis of the output-layer weights indicated that our framework displayed a structured bimodal distribution after pre-training (Fig. 17a). This pattern is a direct reflection of the multivariate dependencies encoded in the LUT. Notably, this structured distribution remained stable even after fine-tuning. In contrast, the data-driven MLP evolved from an unstructured initial state (Kaiming initialization) to a partially organized distribution after extensive training (Fig. 17b). This stability in the parameter space corresponds to the validation loss trajectories illustrated in Fig. 10, suggesting that the optimization acts as a local refinement within a nearly optimal basin, which accounts for the accelerated convergence of our framework.

To explicitly connect the findings from the parameter space to model behavior, we analyze the Pearson correlation coefficients between the predicted CHF and each input feature across different modeling stages, as illustrated in Fig. 18. The row containing experimental data presents the actual physical trends: there is a negative correlation with pressure (-0.30) and heated length (-0.37), a positive correlation with mass flux ($+0.51$), and weaker influences from other parameters.

The MLP, following its training, effectively captures the dominant correlations, demonstrating its ability in learning from data within the training domain. However, its initial state, achieved through Kaiming initialization, presents predominantly random and frequently unphysical correlations. This indicates that before training, the parameterization does not convey any meaningful thermohydraulic structure.

The LUT-pretrained model, which has not been exposed to any experimental data, effectively captures both the correct directional trends and approximate magnitudes of various correlations. This indicates that pretraining effectively encodes not only individual parameter influences but also their relative importance. When fine-tuning is

applied, it leads to only minor adjustments, resulting in correlations from the fine-tuned model that are nearly identical to the experimental data. This behavior supports the findings regarding weight distribution: pretraining positions the network within a parameter regime that inherently reflects physically consistent dependencies, while fine-tuning primarily adjusts and calibrates these established relationships rather than alters them.

In summary, the hybrid framework remains coupled to the prior's error structure; when these errors become irregular beyond the training domain, extrapolation degrades as the residual becomes difficult to learn. In contrast, the proposed pretraining framework decouples these roles: it learns the fundamental physical trends from abundant synthetic data during pretraining and calibrates the biases with scarce experimental data during fine-tuning.

3.4. Extrapolation mapping for data acquisition guidance

The exceptional pressure-extrapolation capability of the proposed framework offers a valuable tool for optimizing future data-acquisition strategies. To assess out-of-distribution (OOD) reliability in a clear and actionable way, we develop an extrapolation mapping methodology, conceptually adapted from reference [29]. This methodology graphically delineates the areas where the model demonstrates reliable generalization and identifies the regions where additional experimental data is most urgently needed.

This analysis leveraged a proprietary database of CHF data. As shown in Fig. 19, the pressure domain was divided using two distinct binning strategies. The training bins (on the x-axis) comprised consecutive intervals of 3 MPa, while the testing bins (on the y-axis) were divided into finer, non-overlapping segments of 1 MPa. For each training bin i , a dedicated instance of the proposed pre-training framework was trained exclusively using data within that pressure range. Each trained model was then evaluated against each testing bin j . The assessment of extrapolation performance is confined to the off-diagonal elements of the heatmap in Fig. 19. The diagonal elements, shaded in gray,

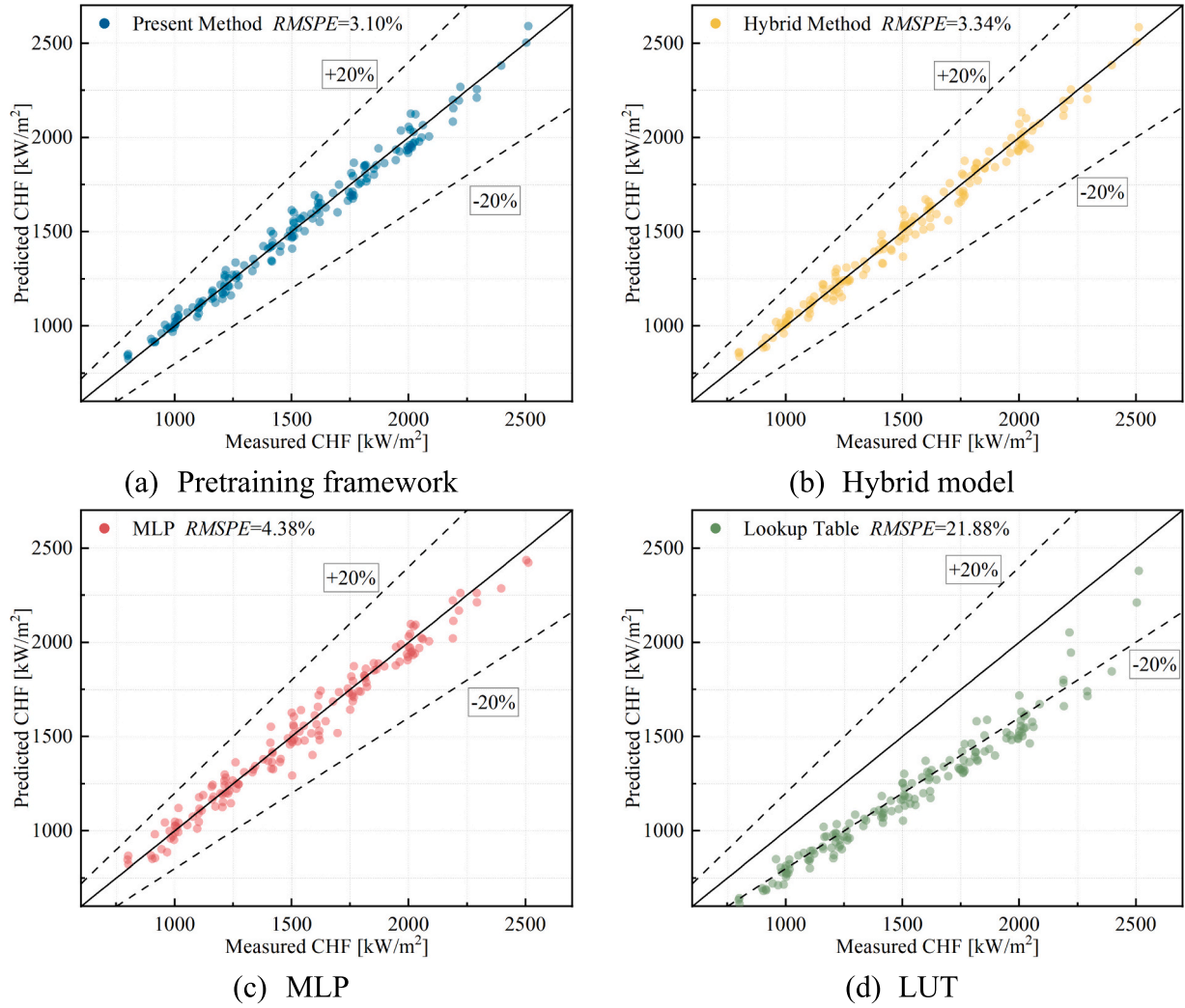


Fig. 13. Scatter plot of predicted versus experimental CHF values for the 5×5 fuel assembly test set.

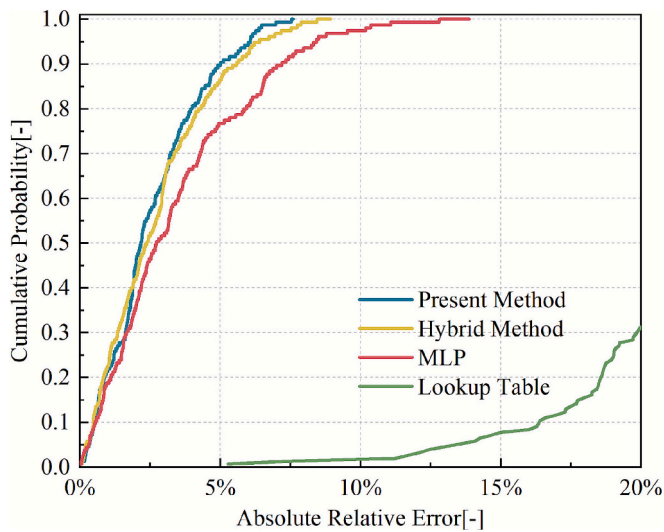


Fig. 14. Cumulative distribution of the absolute relative error for the 5×5 fuel assembly test set.

Table 4

Extrapolation performance metrics on the 5×5 fuel assembly dataset under different pressure cutoff values.

Cutoff Pressure P_{cutoff} [MPa]		10	12	14	16
No. of training data		170	234	293	520
No. of test data		561	481	407	123
Test data R^2	Present Method	0.9560	0.9776	0.9804	0.9836
	Hybrid Method	0.8231	0.9620	0.9686	0.9791
	MLP	0.6741	0.9097	0.9292	0.9749
	Present Method	0.0559	0.0402	0.0378	0.0351
Test data rRMSE	Hybrid Method	0.1120	0.0523	0.0478	0.0396
	MLP	0.1521	0.0807	0.0718	0.0433
	Present Method	4.95 %	3.47 %	3.08 %	2.87 %
	Hybrid Method	10.74 %	4.63 %	3.92 %	3.42 %
Test data MAPE	MLP	12.39 %	6.14 %	6.28 %	3.47 %
	Present Method	86.81 %	98.13 %	99.75 %	100.00 %
	Hybrid Method	52.23 %	91.27 %	92.63 %	98.37 %
	MLP	41.00 %	81.29 %	82.31 %	96.75 %

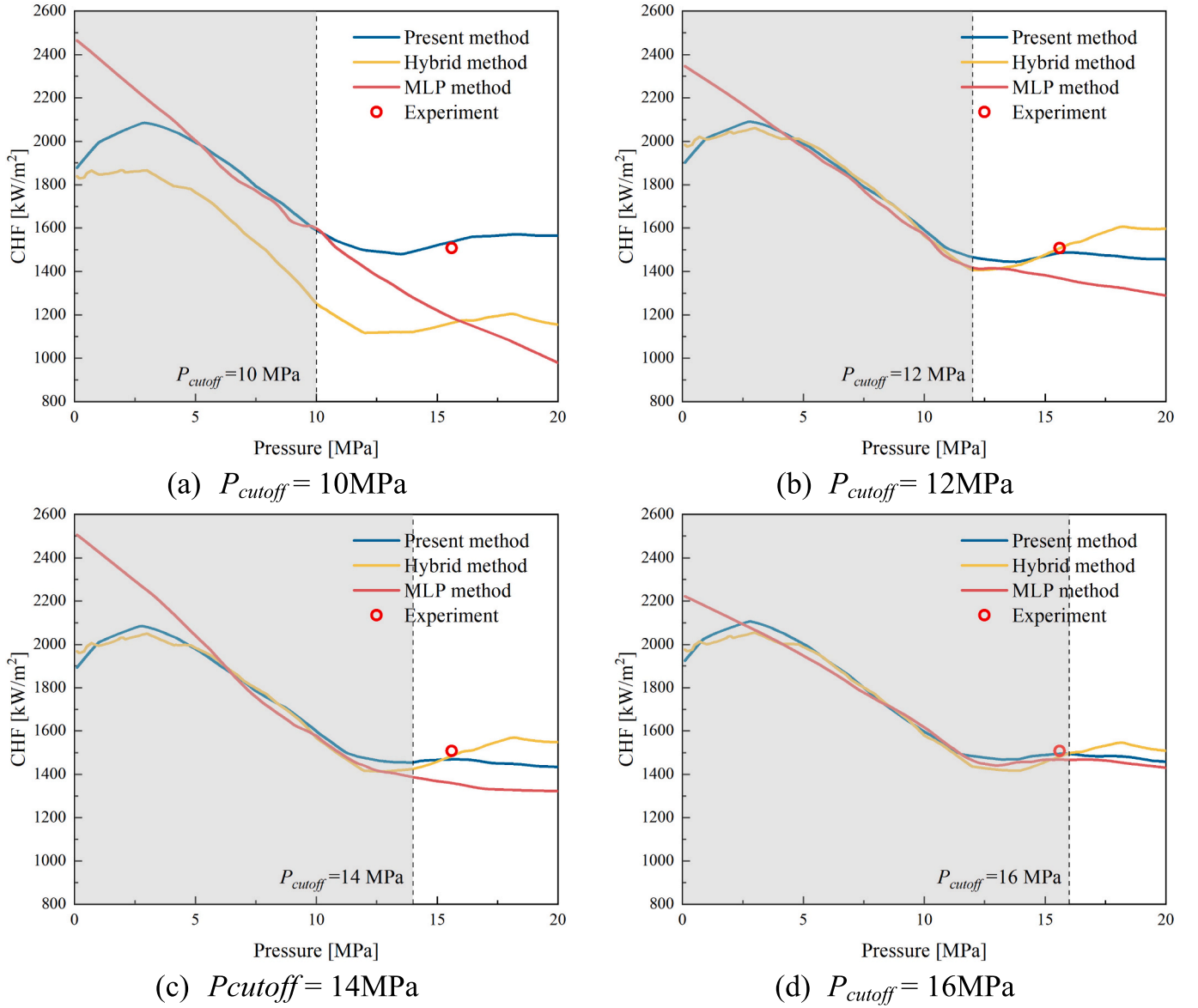


Fig. 15. Comparison of predicted CHF trends over pressure for the 5×5 fuel assembly under different training cutoff pressures.

representing interpolation or near-interpolation conditions, are provided solely as a reference baseline for comparison. The extrapolation capability for each training-testing pair (i, j) was quantified using the rRMSE.

The extrapolation map demonstrated a clear and interpretable structure. Areas near the gray diagonal, where the training and testing pressures were close yet distinct, showed low errors, indicating strong near-range extrapolation. More importantly, the off-diagonal patterns provide insights into broader extrapolation performance. A significant finding is that models trained on mid-to-high pressure ranges (e.g., 9–12 MPa or 16–19 MPa) can effectively generalize to unseen intermediate pressures. This suggests that to predict CHF at 15 MPa, which is a potential target for new systems, existing data from surrounding pressure ranges are generally sufficient. As a result, new experiments at 15 MPa were given lower priority, as they would not greatly improve predictive capabilities for this specific condition.

Conversely, the map identifies regions with high extrapolation uncertainty. The upper left quadrant shows that models trained exclusively on low-pressure data (e.g., 0.0–4.5 MPa) performed poorly when extrapolated to high pressures. Notably, the horizontal band corresponding to the 3.5–5.5 MPa test interval exhibited consistently high

rRMSE values (>0.25), irrespective of the training bin. This indicates a fundamental gap in the current database; the physical mechanisms governing CHF in this transitional pressure regime are not adequately captured by models based on data from other regions. Therefore, the 3.5–5.5 MPa range should be the highest-priority domain for future experimental campaigns. Acquiring data within this window is essential for anchoring the model's predictions and improving its overall generalizability.

In summary, this extrapolation mapping transforms the predictive model from a passive tool to an active guide for strategic resource allocation. It provides a quantitative, evidence-based strategy for experimental design, enabling researchers to focus on collecting data in regions that will most effectively reduce model uncertainty and expand its reliable operational domain.

4. Conclusions

This study proposes a novel pretraining framework for predicting CHF, thereby establishing a paradigm shift from conventional deviation-correction methods to a pretrained initialization strategy. The framework was thoroughly benchmarked against a standard MLP and a state-

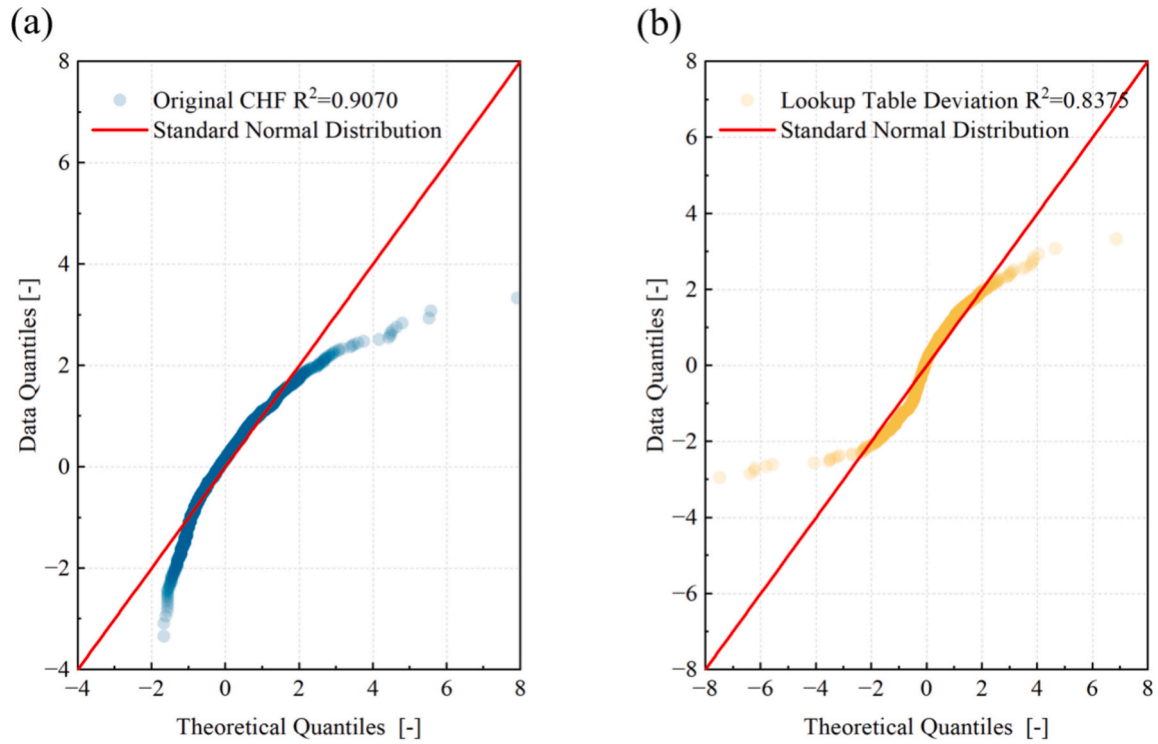


Fig. 16. Quantile-quantile (Q-Q) plots of the (a) experimental CHF database and (b) the deviation between the Lookup Table predictions and experimental values, each compared to a normal distribution.

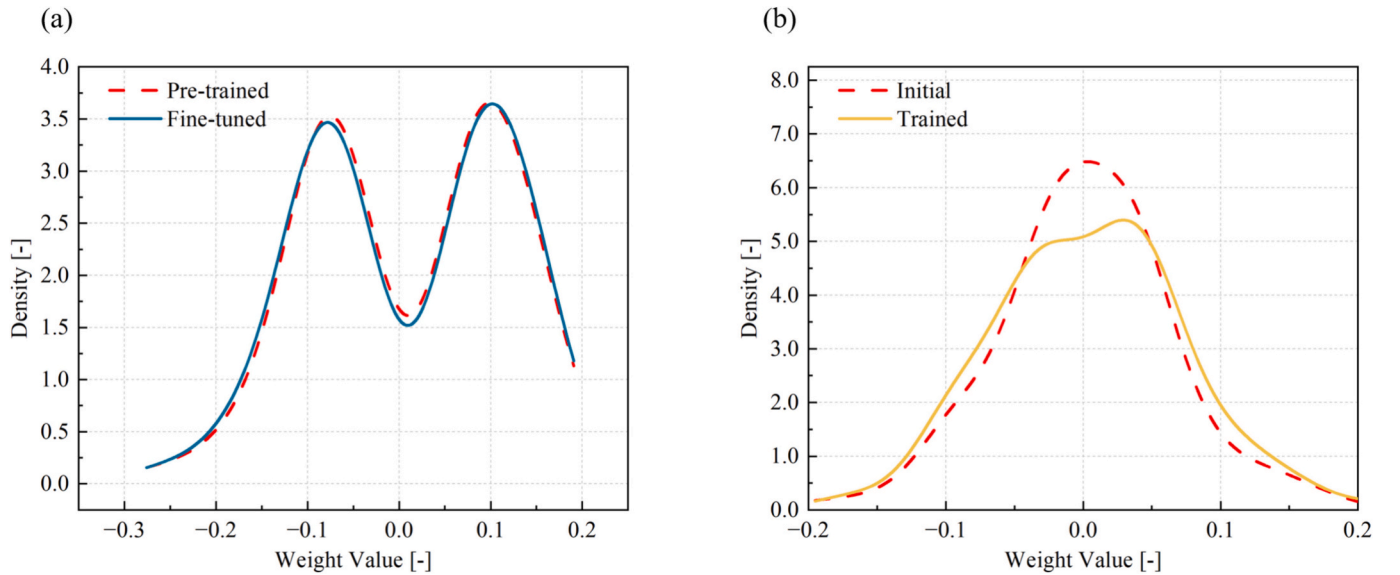


Fig. 17. Comparison of output-layer weight distributions for (a) the proposed framework and (b) the standard MLP.

of-the-art hybrid model on a consolidated database of 1637 tube flow CHF data points and an additional 768 proprietary measurements from 5×5 fuel assemblies. The key findings and implications of this work are summarized as follows:

1. The proposed framework exhibited enhanced overall performance, exhibiting accelerated convergence and significantly outperforming both the MLP and hybrid models in terms of interpolation accuracy and pressure-extrapolation capability across both tube flow and fuel assembly datasets. Even in the most challenging extrapolation scenarios, the proposed model delivered robust and physically

consistent predictions, while the baseline models exhibited substantial performance degradation.

2. The hybrid framework suffers from a foundational flaw: it learns the prior model's errors rather than its physics. Our analysis shows these errors are non-systematic and harder to model than the true CHF phenomenon. Consequently, during extrapolation into regimes devoid of training data, the model lacks the necessary information to learn a meaningful correction to the prior's inherent inconsistencies, resulting in the catastrophic failures observed in our pressure-extrapolation tests.

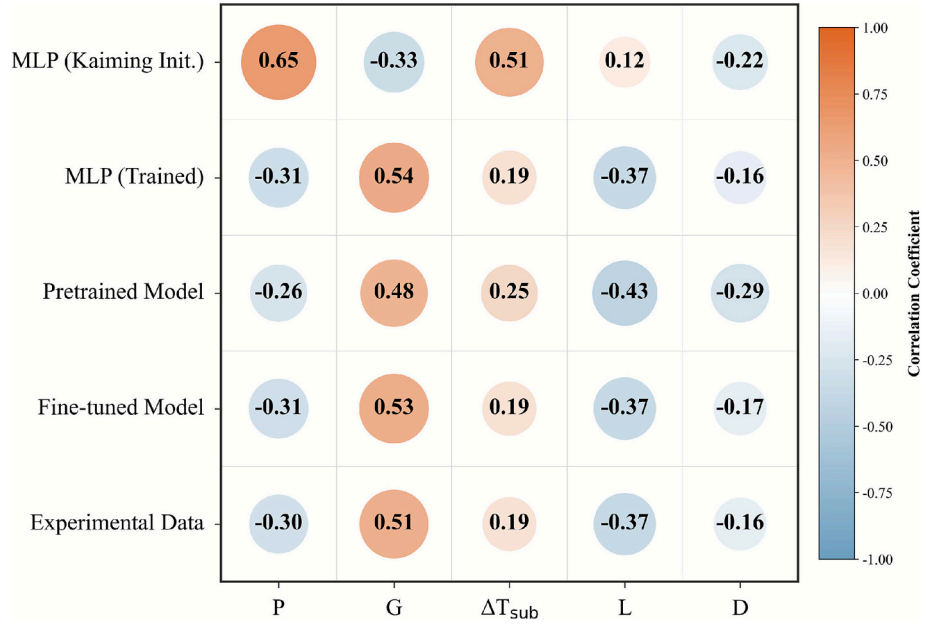


Fig. 18. Comparison of input feature-CHF correlations for different models and experimental data.

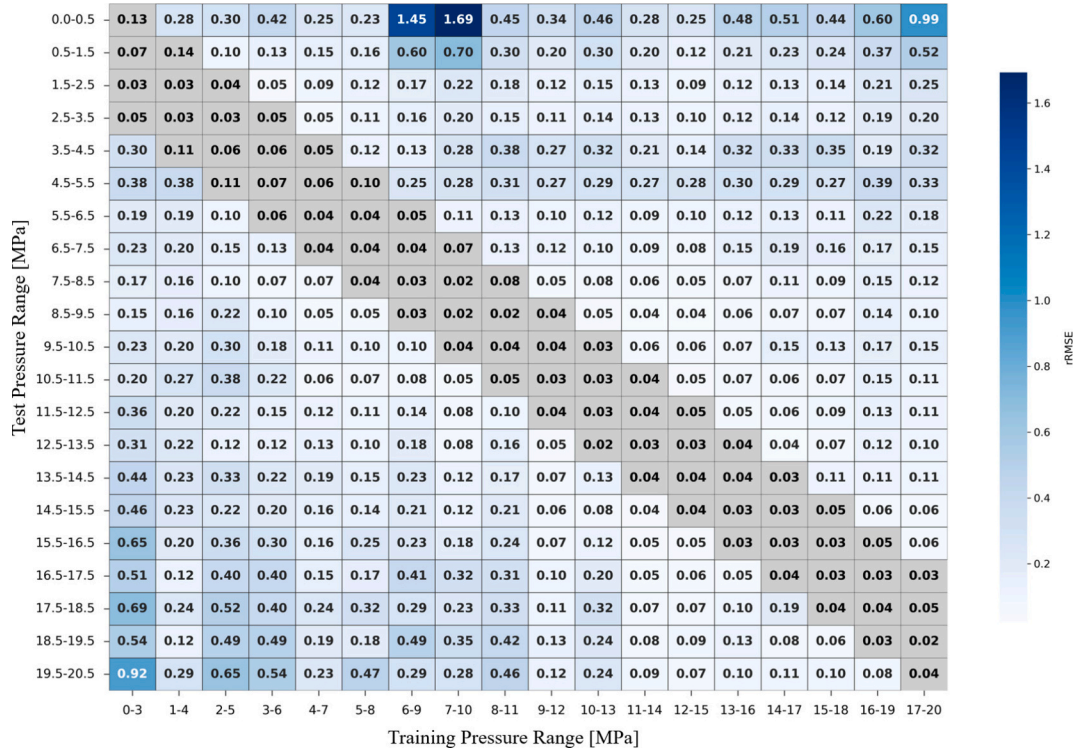


Fig. 19. Window-type extrapolation performance map of the proposed model.

3. The framework demonstrated remarkable generalization capability to complex geometries, as evidenced by its performance on the 5×5 fuel assembly dataset. This confirms that pretraining on the LUT, while not a first-principles model, provides a domain-relevant initialization that is highly effective for transfer learning, even when the target domain involves different CHF mechanisms.
4. Interpretability analysis indicated that the pretraining phase successfully encoded fundamental thermohydraulic dependencies directly into the network's parameter space. This resulted in a structured and stable weight distribution, establishing a robust,

physics-consistent initial state. The subsequent fine-tuning then served as a local adjustment of this prior knowledge, which contributed to the framework's improved generalization and faster convergence.

5. The window-type extrapolation mapping provided a quantitative and actionable tool for guiding future experimental efforts. It identified specific pressure regimes as critical targets for new data acquisition and highlighted regions where existing data provides adequate coverage for reliable prediction.

We emphasize that the proposed transfer learning framework does not eliminate the intrinsic data dependence of CHF modeling; rather, it redistributes this dependence by leveraging an established and well-validated knowledge base to reduce the amount of additional, domain-specific experimental data required for accurate prediction. Despite these advancements, several promising directions remain for future work to further enhance the generality and physical fidelity of CHF prediction models:

1. The current database and models lack descriptors of heated surface properties, such as roughness, wettability, and nucleation site density, which are known to significantly influence bubble dynamics and CHF mechanisms. It is noteworthy that the experimental data utilized in this study, as well as the majority of the underlying data for the 2006 LUT, were obtained from metallic surfaces with industrial-grade surface finish. Future models should incorporate surface feature embeddings, and corresponding experiments must be designed to quantify these effects.
2. To comprehensively assess the transferability of the proposed framework, proprietary CHF data for other industrially relevant complex geometries should be made accessible under appropriate data governance and confidentiality protocols. A concerted effort from both industry and academia is strongly encouraged to accelerate this transition and build more generalizable next-generation safety tools.
3. The proposed pretraining framework is a general methodology. Future research could explore its application with other foundational physical models and extend it to other challenging domains in thermal-hydraulics, where physics-based guidance is critical but data is scarce.
4. The present study is primarily methodological and empirical in nature. While the proposed framework illustrates how LUT-based pretraining guides the optimization process and enhances extrapolation performance, it does not provide a rigorous theoretical analysis of generalization guarantees or convergence properties. Future research will focus on developing a deeper theoretical understanding of these mechanisms.

Nomenclature

Latin symbols

D	Tube diameter
G	Mass flux
L	Heated length
P	Pressure
q	Heat flux
R^2	Coefficient of determination
T	Temperature
ΔT_{sub}	Inlet subcooling
x	Local equilibrium quality

Greek symbols

α	Relaxation factor for iterative CHF calculation
ε	Convergence tolerance for model training
Δ	Increment symbol

Subscripts

cutoff	Cutoff pressure for data partitioning
LUT	Denotes value from 2006 CHF Lookup Table
sub	Subcooling

Abbreviations

CHF	Critical Heat Flux
CNN	Convolutional Neural Network
CFD	Computational Fluid Dynamics
CDF	Cumulative Distribution Function
LUT	Lookup Table
MLP	Multilayer Perceptron
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
OOD	Out-of-Distribution
PCA	Principal Component Analysis
RMSPE	Root Mean Square Percentage Error
rRMSE	Relative Root Mean Square Error

CRedit authorship contribution statement

Shihao Yang: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Xiaoling Li:** Writing – original draft, Visualization, Investigation. **Bo Ren:** Software, Methodology, Formal analysis. **Lixin Yang:** Supervision, Methodology, Funding acquisition. **Chong Chen:** Software, Funding acquisition, Formal analysis. **Qi Lu:** Visualization, Resources, Project administration. **Zonglan Wei:** Visualization, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2025.129451>.

Data availability

Data will be made available on request.

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